

Membranous Nephropathy

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DEFINITION

Membranous nephropathy (MN) is an immune complex disease in which deposits of immunoglobulin (Ig) G and complement components develop predominantly or exclusively beneath podocytes on the subepithelial surface of the glomerular capillary wall. Podocyte injury resulting from the immune deposits increases glomerular permeability, which results in proteinuria and potentially in nephrotic syndrome.^{1,2} *Primary* (formerly called *idiopathic*) MN is an organ-specific autoimmune disease in which autoantibodies target an intrinsic podocyte antigen. It accounts for about 75% to 80% of patients with MN and typically occurs in the absence of any identifiable initiating event. It is the most common cause of primary nephrotic syndrome in older (>60 years) White adults, but the age range is broad and patients may present for the first time as teenagers.³ Various conditions have been identified in association with MN, some of which are likely to be causal, and are known as *secondary* MN (Table 21.1). The term “membranous” refers to thickening of the glomerular capillary wall on light microscopy of a kidney biopsy. Currently MN is more often defined by immunofluorescence and electron microscopy showing the pathognomonic diffuse, finely granular or electron-dense immune deposits in the subepithelial space. Consequently, MN is a pathologic diagnosis made in patients with proteinuria whose glomeruli exhibit these immune deposits without associated hypercellularity or inflammatory changes.

ETIOLOGY AND PATHOGENESIS

Experimental Membranous Nephropathy

Much of what we know about the pathogenesis of MN derives from observations in animal models.⁴ Studies of the Heymann nephritis model of MN in rats established that the subepithelial immune deposits form in situ when circulating antibodies bind to an intrinsic (“fixed”) antigen in the glomerular capillary wall. The antigen in rats was identified as *megalyn*, a large transmembrane receptor of the low-density lipoprotein receptor family expressed on the basal surface of rat podocytes. Binding of circulating antimegalyn antibodies induces capping and shedding of the antigen-antibody complexes, where they bind to the underlying glomerular basement membrane (GBM), resist degradation, and persist for weeks or months as immune deposits characteristic of MN (Fig. 21.1A). In this model, proteinuria is caused by antibodies in the deposits that overcome local complement regulatory mechanisms and activate complement in situ. Sublethal podocyte injury induced by the complement membrane attack complex C5b-9 triggers a cascade of changes, including oxidative injury, calcium influx, activation of cytosolic phospholipase A₂, production of arachidonic acid metabolites and cytokines, endoplasmic reticulum stress, DNA damage, and alterations in the ubiquitin-proteasome system⁴

(Fig. 21.2). Podocyte foot process effacement results from collapse of the actin cytoskeleton and loss of cell-GBM adhesion complexes, and the loss and displacement of slit diaphragms is associated with the onset of severe, nonselective proteinuria. Podocyte injury also leads to the production of new extracellular matrix proteins that are laid down around the immune deposits, giving rise to the characteristic “spikes” and GBM thickening.

Another mechanism of subepithelial immune deposit formation involves planted antigens² (Fig. 21.1B). This is best exemplified by animal models immunized with cationized bovine serum albumin (cBSA). The cBSA binds to negatively charged residues in the GBM, where it serves as a target for circulating anti-BSA antibodies. As in the Heymann nephritis model, podocyte injury and proteinuria result from local complement activation.

Human Membranous Nephropathy

The first demonstration that circulating antibodies reactive with an intrinsic podocyte antigen might be involved in human MN (see Fig. 21.1) was provided by an unusual case of antenatal MN induced by the transplacental passage of alloantibodies to podocyte neutral endopeptidase (NEP).⁵ The mother of the affected child was deficient in NEP and had been immunized during a previous pregnancy and, like subsequent cases, produced complement-fixing anti-NEP alloantibodies. Although these anti-NEP alloantibodies are distinct from the autoantibodies responsible for adult primary MN, other alloantibodies probably explain the development of de novo MN after kidney transplantation and MN in the setting of chronic graft-versus-host disease after allogeneic hematopoietic stem cell transplantation (see Table 21.1).

Since this initial demonstration of a podocyte-expressed antigen in human MN, several additional target antigens have been identified. The predominant autoimmune system responsible for primary MN is that associated with autoantibodies directed at the M-type phospholipase A₂ receptor (PLA₂R) on podocytes.^{4,6} Circulating anti-PLA₂R antibodies are detectable in the serum of 75% to 80% of patients with primary MN from all ethnic groups and are rarely found in secondary MN.^{2,7} The antibodies are predominantly IgG4, and the antigen (PLA₂R) and antibodies (anti-PLA₂R) colocalize in the immune deposits in primary (but not in secondary) MN.^{8,9} These IgG4 anti-PLA₂R autoantibodies are able to initiate complement activation through the lectin pathway,¹⁰ although other subclasses of anti-PLA₂R and other complement pathways are likely also involved. PLA₂R is a transmembrane protein of the mannose receptor family¹¹ (Fig. 21.3) that undergoes constitutive endocytosis between the plasma membrane and internal compartments; its role in podocytes is unknown.

In the 20% or more cases of primary MN not associated with antibodies to PLA₂R, a number of distinct target antigens have been identified that now define subsets of MN, some of which have distinct

TABLE 21.1 Classification of Conditions and Agents Associated With Membranous Nephropathy**Primary**PLA₂R-associated (70%–80%)

Other antigens: NELL1 (5%–10%), THSD7A (1%–5%), Sema3B, PCDH7, HTRA1

Secondary**Common****Uncommon**

Autoimmune diseases

Class V lupus nephritis
Can be associated with staining for EXT1/EXT2, NCAM1, TGFBR3 in depositsRheumatoid arthritis
Autoimmune thyroid disease
IgG4-related systemic disease
Crescentic glomerulonephritis (anti-GBM or ANCA-associated)
can co-occur with MN

Infections

Hepatitis B

HCV
HIV
Syphilis
Schistosomiasis

Malignancy

Solid tumors (prostate, breast, lung, colon, stomach, kidney)

Non-Hodgkin lymphoma
CLL
Melanoma

Drugs or toxins

NSAIDs and COX-2 inhibitors

Mercury-containing compounds
Gold salts
D-Penicillamine, bucillamine

Miscellaneous

Sarcoidosis
Anti-cationic bovine serum albumin**Alloimmune**

Graft-versus-host disease following hematopoietic stem cell transplantation

De novo membranous nephropathy in kidney allograft

Fetomaternal alloimmunization to neutral endopeptidase

ANCA, Antineutrophil cytoplasmic antibody; CLL, chronic lymphocytic leukemia; COX-2, cyclooxygenase-2; EXT1/EXT2, exostosin 1/exostosin 2; GBM, glomerular basement membrane; HCV, hepatitis C virus; HIV, human immunodeficiency virus; HTRA1, serine protease HTRA1; Ig, immunoglobulin; MN, membranous nephropathy; NCAM1, neural cell adhesion molecule 1; NELL1, neural EGFL like 1; NSAID, nonsteroidal antiinflammatory drug; PCDH7, protocadherin 7; PLA₂R, phospholipase A2 receptor; Sema3B, semaphorin 3B; TGFBR3, transforming growth factor beta receptor 3; THSD7A, thrombospondin type-1 domain-containing 7A.

phenotypes or associations. Antibodies to thrombospondin type 1 domain-containing 7A (THSD7A)¹² account for about 5% of cases of primary MN in Western countries but appear to be more prevalent in Japanese patients with primary MN.¹³ THSD7A is localized on the basal surface of podocytes and, like PLA₂R, it redistributes to form the subepithelial immune deposits. Neural EGFL-like 1 (NELL1) is the target of IgG1-predominant autoantibodies, and up to one-third of these MN cases may be associated with malignancy.^{14,15} Antibodies to semaphorin 3B can be found in some early childhood cases of MN but can rarely occur in adults as well.¹⁶ In all of these subsets of MN, the target antigen becomes localized within the immune deposits, allowing pathologists to query the tissue by immunostaining.

Reactivity to certain intracellular antigens, including aldose reductase, superoxide dismutase 2, and alpha-enolase, has been detected in primary MN and may contribute to the progression of podocyte injury. The best evidence of a planted antigen mechanism is that described in children with MN who have been exposed to cBSA, presumably in bottled milk. In such cases, the cBSA localizes in the GBM, where it forms complexes with circulating anti-BSA.¹⁷ Planted antigens may also be responsible for immune deposition in class V (membranous) lupus nephritis and hepatitis B virus (HBV)-associated MN.

In all forms of MN, injury to the podocyte and glomerular filtration barrier continues as long as there is formation of subepithelial antibody-antigen complexes that can activate complement. In primary autoimmune forms, the persistence of circulating autoantibodies is the main factor causing continued disease activity. Only after

these autoantibodies decline and disappear (owing to a spontaneous or treatment-induced remission) can a clinical remission be attained. When immunologic remission occurs before extensive podocyte loss and GBM remodeling, complete recovery is possible, but proteinuria may persist for months. Once there is extensive podocyte loss, proteinuria persists despite immunologic remission, and glomerular sclerosis, tubular atrophy, and interstitial fibrosis may ensue.

EPIDEMIOLOGY AND GENETICS

MN may occur at any age and in all ethnic groups, but primary MN is twice as common in men than in women and is rare in children. Primary MN has its peak incidence during the fourth and fifth decades of life (Box 21.1). MN in childhood is more often secondary (e.g., caused by HBV). Primary MN is the most common cause of nephrotic syndrome in nondiabetic White adults. Estimated annual incidence is 8 to 10 cases per 1 million population in Western countries. Reported variation in incidence may reflect specific country/physician indication for kidney biopsy but could also represent real differences related to socioeconomic status, ethnicity, or environment in particular air pollution. Although association with certain human leukocyte antigen (HLA) class II immune response genes indicated a genetic predisposition, primary MN is typically not a familial disease. Studies from Korea and Taiwan suggested the presence of PLA₂R1 risk variants that were nonsynonymous single-nucleotide polymorphisms (SNPs)^{18,19} in the first and seventh C-type lectin-like domains (CTLDS) of PLA₂R (Fig. 21.3). A subsequent

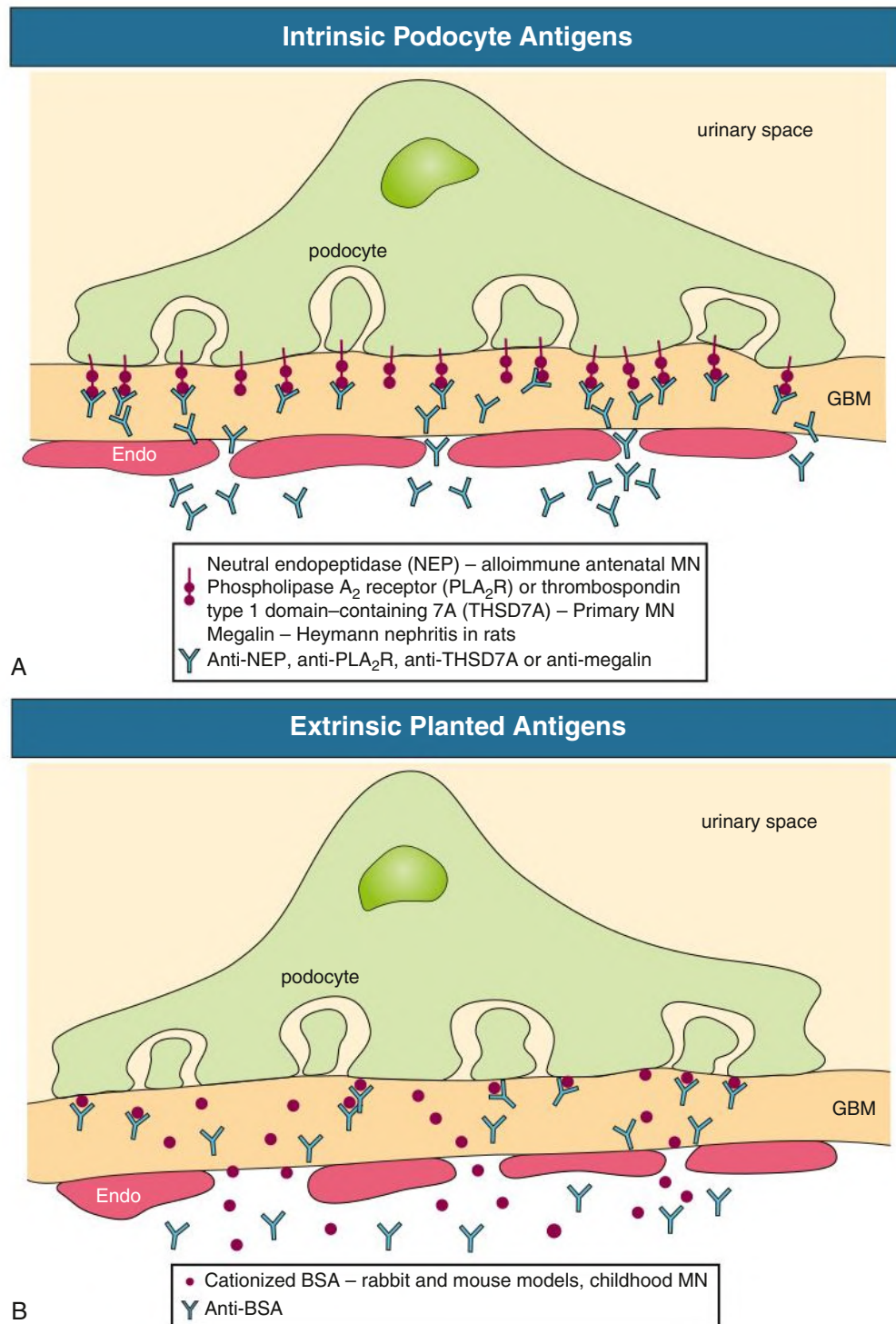


Fig. 21.1 Mechanisms of In Situ Immune Complex Formation in Experimental and Human Membranous Nephropathy (MN). (A) Circulating antibodies may cross the glomerular capillary wall and bind to podocyte antigens exposed on the foot process, as in Heymann nephritis in rats (megalin), alloimmune MN (neutral endopeptidase), and primary MN (PLA₂R or THSD7A). (B) Certain extrinsic antigens such as cationized bovine serum albumin (BSA) may bind to sites in the GBM and serve as planted antigens and form deposits with circulating antibody. *Endo*, Glomerular endothelial cell; *GBM*, glomerular basement membrane.

European genome-wide association study revealed strong associations with a noncoding SNP in *PLA2R1* and another in *HLA-DQA1*, a member of HLA class II that includes isoforms that predispose carriers to autoimmunity.²⁰ The odds ratio of MN was almost 80 in individuals who were homozygous for both *HLA-DQA1* and *PLA2R1* variants. No

unique coding variants have been found in *PLA2R1*, and the risk allele in *PLA2R1* is proposed to lie in a regulatory region that may increase renal expression of the protein.²¹

Higher-resolution analysis of risk variants with MN in the Chinese population suggest that specific HLA molecules may be altered

Postulated Mechanisms of Injury in Experimental Membranous Nephropathy

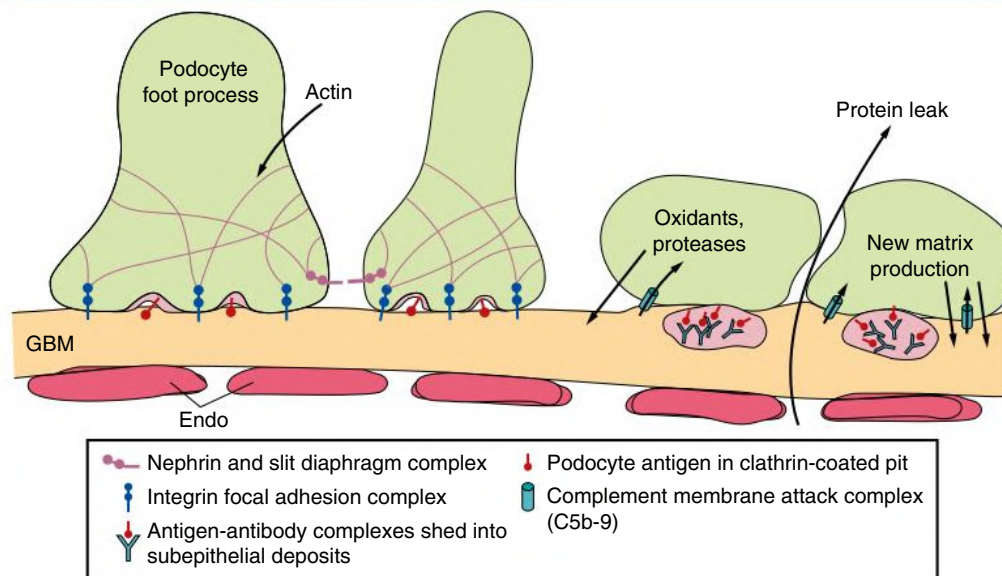


Fig. 21.2 Postulated Mechanisms of Injury in Experimental Membranous Nephropathy. Antibodies against a podocyte antigen in clathrin-coated pits on the foot processes (left podocyte) form complexes that are shed to form deposits in the subepithelial space (right podocytes) and induce complement activation leading to formation of C5b-9. Insertion of C5b-9 is insufficient to cause lysis but stimulates the podocyte to release a host of inflammatory mediators. Disruption of the actin cytoskeleton causes altered cell-matrix adhesion and loss or displacement of slit diaphragms, leading to foot process effacement and loss of the filtration barrier to protein. New matrix production by the damaged podocytes expands the glomerular basement membrane (GBM) between and around the deposits. Endo, Glomerular endothelial cells.

BOX 21.1 Clinical Features of Membranous Nephropathy

- Rare in children: <5% of total cases of nephrotic syndrome
- Common in adults: 15%–50% of total cases of nephrotic syndrome, depending on age; increasing frequency after age 40 years
- Males > females in primary forms; SLE-associated forms may be female predominant
- Can occur in all racial and ethnic groups
- Nephrotic syndrome in 60%–70% of cases
- Normal or mildly elevated blood pressure at presentation
- Urinary sediment reflective of lipiduria^a; trace hematuria possible, but no RBC casts
- Nonselective proteinuria
- Tendency to thromboembolic disease^b
- Associations to suggest possible secondary MN: infection, drugs, neoplasia, SLE

^aFatty casts, oval fat bodies, cholesterol crystals.

^bDeep venous thrombosis, renal vein thrombosis, and pulmonary embolism.

MN, Membranous nephropathy; RBC, red blood cell; SLE, systemic lupus erythematosus.

specifically within the peptide binding groove,²² which may confer the ability to present regions of the PLA₂R protein more effectively to the immune system. PLA₂R1 gene variants may also contribute to the severity or likelihood of MN progression. Alternatively, the interaction of PLA₂R1 with the autoimmune predisposition conferred by HLA

class II may set the stage for an external trigger to initiate MN.^{1,2} An example of such a trigger might be exposure to environmental pollutants.²³ A second genome-wide association confirmed the risk variants within PLA₂R1 and the HLA region and identified additional risk loci in IRF4 and NFKB1,²¹ although the implications of these findings are not yet known.

CLINICAL AND SEROLOGIC MANIFESTATIONS

A majority of patients (70%–80%) with MN present with the nephrotic syndrome. The remaining patients present with asymptomatic subnephrotic, nonselective proteinuria (<3.5 g/24 h) that may be present for months or even years before the diagnosis is made.²⁴ Microscopic hematuria is common (30%–40%), but red blood cell (RBC) casts are rare and suggest a different or coexistent proliferative glomerular pathologic process. In primary MN, serologic tests for anti-PLA₂R are positive in 75% to 80% of cases,^{1,2} whereas serum complement levels are normal despite evidence of intraglomerular complement activation, and serologic markers (e.g., antinuclear antibodies, antineutrophil cytoplasmic antibodies [ANCA], rheumatoid factor) are normal or absent. At diagnosis, 10% to 20% of MN patients have hypertension and kidney function is usually normal, with only a small fraction (<10%) presenting with kidney impairment (Table 21.2). Hypertension and reduced GFR are more common at presentation in older patients; this may reflect tubulointerstitial and vascular changes on biopsy independent of the severity of the MN.²⁵ Complications of the nephrotic syndrome including dyslipidemia, infections, thromboembolic events, and cardiovascular events may be present early in the course of the disease (see later).

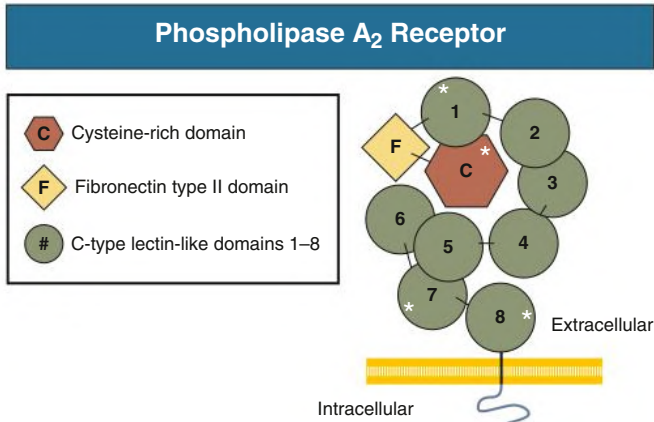


Fig. 21.3 Structural Diagram of Phospholipase A₂ Receptor (PLA₂R), the Major Podocyte Antigen Targeted in Membranous Nephropathy. PLA₂R is composed of 10 extracellular domains, starting with the N-terminal cysteine-rich domain (CysR; red hexagon) and followed by a single fibronectin type II domain (yellow diamond) and eight C-type lectin-like domains (CTLD; green circles). These domains form a compact structure with pH-dependent conformational relationships to each other that likely allow the receptor to release its cargo (such as secreted phospholipase A₂ enzymes) in the more acidic intracellular organelles. A short C-terminal cytoplasmic domain (blue) contains a motif that allows for constitutive endocytosis of the receptor. Autoantibodies to PLA₂R target at least four different regions in the extracellular domain (CysR, CTLD1, CTLD7, and CTLD8; white asterisks). Within CysR, there is an immunodominant epitope of 31 amino acids that is recognized by all individuals with anti-PLA₂R antibodies.³³ (Data from Dong Y, Cao L, Tang H, Shi X, He Y. Structure of human M-type phospholipase A2 receptor revealed by cryo-electron microscopy. *J Mol Biol.* 2017;429[24]:3825–3835; and Yu B, Hu Z, Kong D, Cheng C, He Y. Crystal structure of the CTLD7 domain of human M-type phospholipase A2 receptor. *J Struct Biol.* 2019;207[3]:295–300.)

PATHOLOGY

The earliest pathologic feature of MN is the formation of subepithelial immune complexes of IgG and complement along the outer surface of the capillary wall in which glomeruli may appear histologically normal if only light microscopy is performed. Subsequent changes involve the podocyte, deposition of new extracellular matrix material between and around the immune deposits, thickening of the GBM (membranous change), and in some cases, focal glomerulosclerosis, tubular atrophy, and interstitial fibrosis.

Light Microscopy

In the earliest stages of MN, the glomeruli and interstitium appear normal on light microscopy, and the diagnosis is made by immunohistology and electron microscopy (Fig. 21.4A). The next stage of MN involves a homogeneous thickening of the capillary wall, seen with light microscopy in sections stained with hematoxylin and eosin or with periodic acid–Schiff (PAS) reagent (Fig. 21.4B). On silver methenamine staining, early projections of the GBM between deposits may be detected in a characteristic spike-like configuration (see Fig. 21.4C). Later, lucencies may develop in the GBM as immune deposits are resorbed, resulting in craters within the thickened GBM.

Glomerular leukocyte infiltration and hypercellularity are absent, probably because chemotactic products of complement activation follow filtration forces into the urinary space, and the intervening GBM prevents immune adherence mechanisms from being operative.

The podocyte response to this form of injury includes effacement of foot processes visible only by electron microscopy. There typically are no visible endothelial cell abnormalities. The presence of significant

TABLE 21.2 Diagnosis and Management of Patients With Membranous Nephropathy

Patient Groups	Test
All patients	Vital signs, especially blood pressure and weight Kidney function (serum creatinine and eGFR) Urine analysis Urine protein excretion (24-hour urine or urine protein/creatinine ratio) Serum albumin Serum cholesterol, including LDL cholesterol Anti-PLA ₂ R Kidney biopsy ^a
Associated disease	Hepatitis B (HBs antigen) Hepatitis C (HCV antibody) ANA, anti-double-stranded DNA (hallmark of systemic lupus erythematosus) Complement C3, C4 (usually normal in primary MN)
Select Patients	
With suspected thromboembolic events, flank pain, hematuria, acute kidney injury	Renal venous Doppler ultrasound Contrast CT, MRI
With sudden decrease in kidney function, development of active urine sediment	Anti-GBM antibody ANCA Assess for interstitial nephritis
Suggestive symptoms or age >50 years	Diagnostic testing for cancer (see text)

^aKidney biopsy may not be necessary in certain situations if anti-PLA₂R seropositivity is detected in an appropriate clinical setting.

ANA, Antinuclear antibody; ANCA, antineutrophil cytoplasmic antibody; Anti-PLA₂R, anti-phospholipase A₂ receptor antibody; CT, computed tomography; eGFR, estimated glomerular filtration rate; GBM, glomerular basement membrane; HCV, hepatitis C virus; LDL-HDL, low-density/high-density lipoprotein; MN, membranous nephropathy; MRI, magnetic resonance imaging.

mesangial hypercellularity suggests immune deposit formation in the mesangium and is more consistent with a secondary MN, such as class V lupus nephritis (see Chapter 27). In some patients with heavy proteinuria and progressive disease, glomeruli exhibit reduced podocyte numbers and areas of focal sclerosis that resemble secondary focal segmental glomerulosclerosis (FSGS; see Chapter 19). These patients often have a more rapidly progressive course and a poor response to therapy. These sclerotic lesions may be a consequence of glomerular hypertrophy accompanied by an inability of terminally differentiated podocytes to proliferate leading to areas of denuded GBM, attachment to Bowman’s capsule, and subsequent capillary collapse. As in all glomerular diseases, tubulointerstitial injury is common, and its severity correlates with both long-term kidney function and the level of proteinuria.

Immunohistology

Granular glomerular capillary wall staining for IgG in MN is characteristic (Fig. 21.5A). Positive staining for IgG marks the finely granular subepithelial deposits, which are present on the outer surface of all capillary walls. The predominant IgG subclass in many forms of primary MN is IgG4,^{6,12} although IgG1 may predominate in other types of MN.^{15,16}

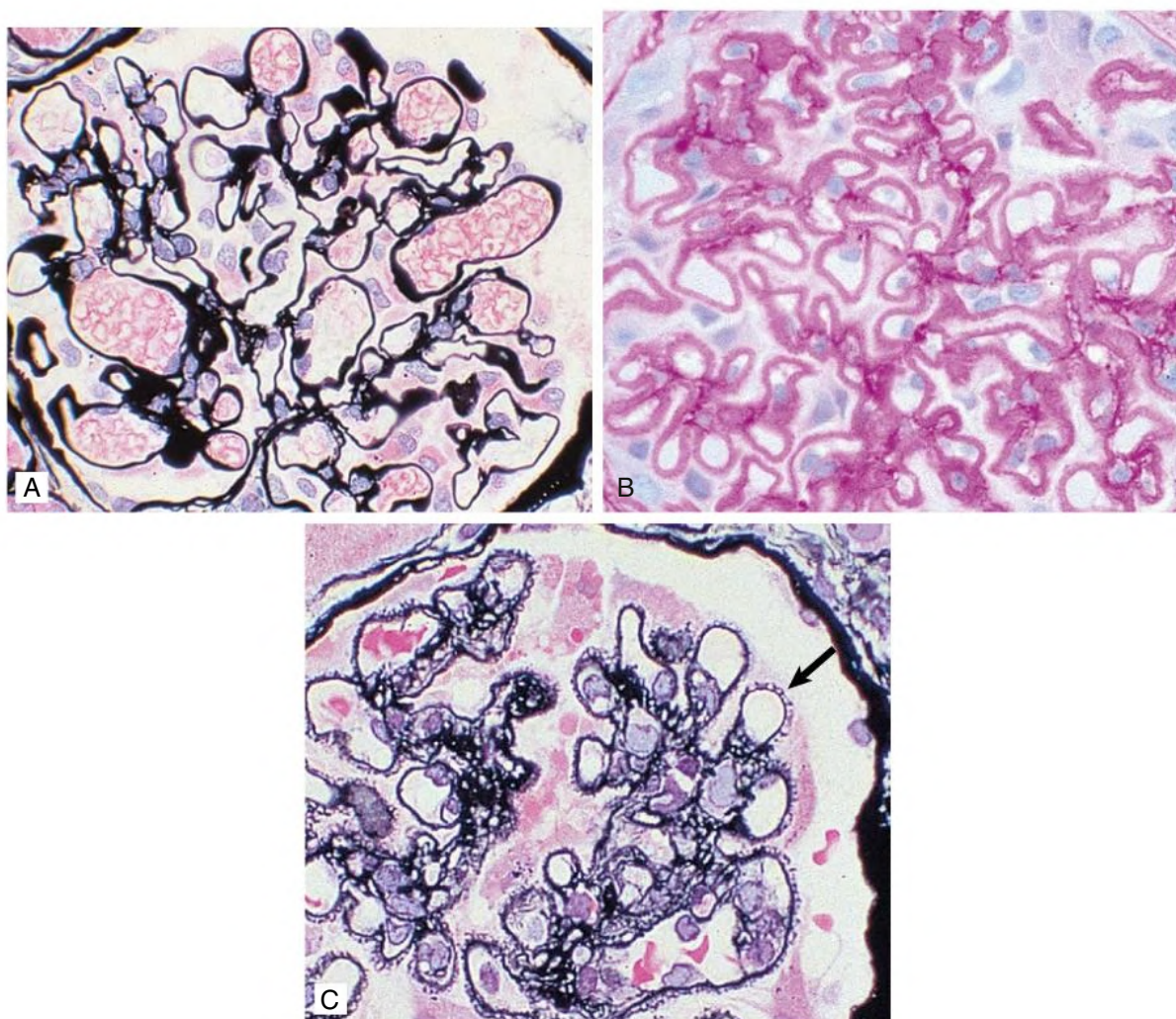


Fig. 21.4 Light Microscopy in Membranous Nephropathy (MN). (A) Early MN. Glomerulus from a patient with severe nephrotic syndrome exhibiting normal architecture and peripheral capillary basement membranes of normal thickness. (B) Morphologically advanced MN. Uniform increase in thickness of glomerular capillary walls throughout the glomerulus with no increase in glomerular cellularity. (C) More morphologically advanced MN, same patient as in (B). Discrete spikes of matrix emanating from outer surface of the basement membrane (*arrow*), indicative of advanced MN. (A and C, Silver methenamine stain, $\times 400$. B, PAS, $\times 400$.) (Courtesy C.E. Alpers.)

Positive staining for IgA, or IgM or significant staining in the glomerular mesangium suggests lupus or other causes of secondary MN.²⁶ Kappa and lambda light chain staining are typically equal, but rare cases of MN due to monoclonal IgG, including anti-PLA₂R, have been reported.²⁶ Complement C3 is present in most cases of active disease and usually reflects staining for C3c, a breakdown product of C3b that itself is rapidly cleared. Consequently, positive C3 staining probably reflects active, ongoing immune deposit formation. When sought, staining for C5b-9 is generally present as well, consistent with the proposed pathogenetic role of C5b-9 in this disease.^{4,27,28} Strong C1q staining is uncommon in primary MN (<20% of cases) but common in lupus-associated MN.²⁹ Positive staining for C4d in the absence of C1q is characteristic of primary MN.³⁰

The identification of target antigens (or specific biomarkers associated with types of MN) now allows immunohistologic subtyping of MN biopsies. These antigens accumulate within immune deposits where they can demonstrate the same fine granular capillary wall pattern as IgG and C3 (because they all reside within the immune deposits). Many pathologists stain the biopsy for PLA₂R and, if negative, may send the slides to be stained for other autoantigens at specialized

centers. PLA₂R staining in the immune deposits in a pattern that colocalizes with IgG is another feature that helps distinguish PLA₂R-associated MN from secondary MN (Fig. 21.5B) (Table 21.3).^{6,8,9}

Electron Microscopy

The finding of subepithelial electron-dense deposits by electron microscopy (EM) parallels IgG staining. In primary MN, immune deposit formation occurs in a subepithelial distribution; subendothelial deposits are not seen and mesangial deposits are rare (Table 21.3). These deposits in early stages of the disease process are homogeneous and may even be confluent in some areas, with overlying podocyte foot process effacement and little change in the underlying GBM (stage I). As the disease persists, basement membrane material is laid down between the deposits and corresponds to the spikes seen on light microscopy and are easily visible by EM (stage II; Fig. 21.6A). Later, the spikes extend and the deposits may become surrounded by new basement membrane-like material (stage III; Fig. 21.6B). In stage IV disease, the basement membrane is overtly thickened, the deposits incorporated in it become more lucent, and the spikes are less apparent

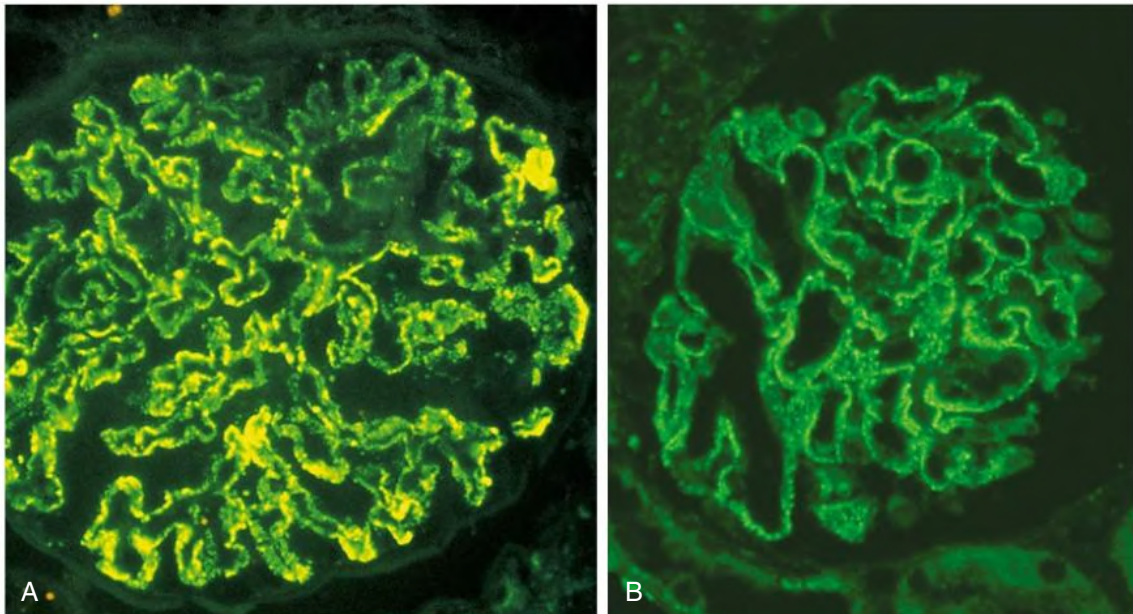


Fig. 21.5 Immunofluorescence in Membranous Nephropathy. (A) Glomerulus with diffuse, finely granular deposits of Immunoglobulin (Ig) G along outer surface of all capillary walls. In primary membranous nephropathy, the antibodies eluted from the glomeruli are reactive with PLA₂R. (B) Glomerulus stained for PLA₂R with diffuse, finely granular deposits of the antigen along outer surface of all capillary walls in a similar distribution as IgG. (Original magnification ×400.) (A, Courtesy C.E. Alpers.)

TABLE 21.3 Histopathologic Features That Help Distinguish Primary From Secondary Membranous Nephropathy

Primary	Secondary
Immunofluorescence Microscopy	
IgA and IgM absent	IgA and/or IgM may be present
IgG4 is predominant or codominant	IgG1, IgG2, or IgG3 predominant
Mesangial Ig staining absent	Mesangial Ig staining may be present
C1q negative or weak	C1q positive
PLA ₂ R positive and colocalizes with IgG	PLA ₂ R negative EXT1/EXT2, NCAM1, or TGFBR3 can colocalize with IgG in MN associated with lupus
Electron Microscopy	
Subepithelial deposits only ± rare mesangial or paramesangial deposits	Subepithelial deposits ± mesangial and subendothelial deposits Tubuloreticular inclusions in endothelial cells

EXT1/EXT2, Exostosin 1/exostosin 2; Ig, immunoglobulin; MN, membranous nephropathy; NCAM1, neural cell adhesion molecule 1; PLA₂R, phospholipase A2 receptor; TGFBR3, transforming growth factor beta receptor 3.

(stage IV; Fig. 21.6C). Although clearly reflecting the duration of disease, these GBM changes do not correlate well with clinical manifestations at time of biopsy or ultimate outcome. The overlying podocyte foot processes are effaced with condensation of the actin cytoskeleton; the filtration slits between the foot processes may be occluded; and in those still open, the slit diaphragms may be displaced or disrupted. Microvillous changes of the podocyte membrane are common, as are

protein reabsorption droplets within podocytes and proximal tubular cells. The presence of tubuloreticular inclusions in the endothelial cells is strongly suggestive of lupus-associated MN, although these may rarely be found in primary MN as well.³¹

Diagnosis and Differential Diagnosis

When the initial presentation includes the nephrotic syndrome, the differential diagnosis includes minimal change disease (MCD), FSGS, the membranoproliferative glomerulonephritis (GN) spectrum of diseases (including C3 glomerulopathies and immune complex GN; see Chapters 22 and 23), amyloidosis, light-chain deposition disease, lupus nephritis, and diabetic nephropathy. In the 20% to 25% of patients whose initial presentation is asymptomatic nonnephrotic proteinuria, the differential is even more extensive. Although clinical clues in proteinuric patients may increase the likelihood of one specific histologic pattern over another, confirmation that MN is the underlying cause of nephrotic syndrome generally requires a kidney biopsy.

In the presence of proteinuria and a positive anti-PLA₂R antibody test measured by both enzyme-linked immunosorbent assay and immunofluorescence assay, estimated GFR (eGFR) greater than 60 mL/min/1.73m², and negative screening for autoimmune and infectious causes as well as malignancy, alternate diagnoses are rarely encountered and a biopsy is not necessary for diagnosis.³² Also, when a biopsy cannot be performed, such as when uninterrupted anticoagulation is required, the diagnosis can be confidently made with a positive serologic test for anti-PLA₂R or anti-THSD7A. On the other hand, negative tests for anti-PLA₂R and anti-THSD7A do not exclude primary MN, as some patients may have positive glomerular staining for PLA₂R or THSD7A despite negative serology early in the course of disease or during remission.¹ In addition, some patients have all the features of primary MN but may have MN associated with one of the other minor antigens noted above.^{1,33}

Secondary MN represents 20% to 30% of all cases (see Table 21.1); the most common causes are systemic lupus, hepatitis B, malignant neoplasms, and drugs. In addition to a careful history and physical

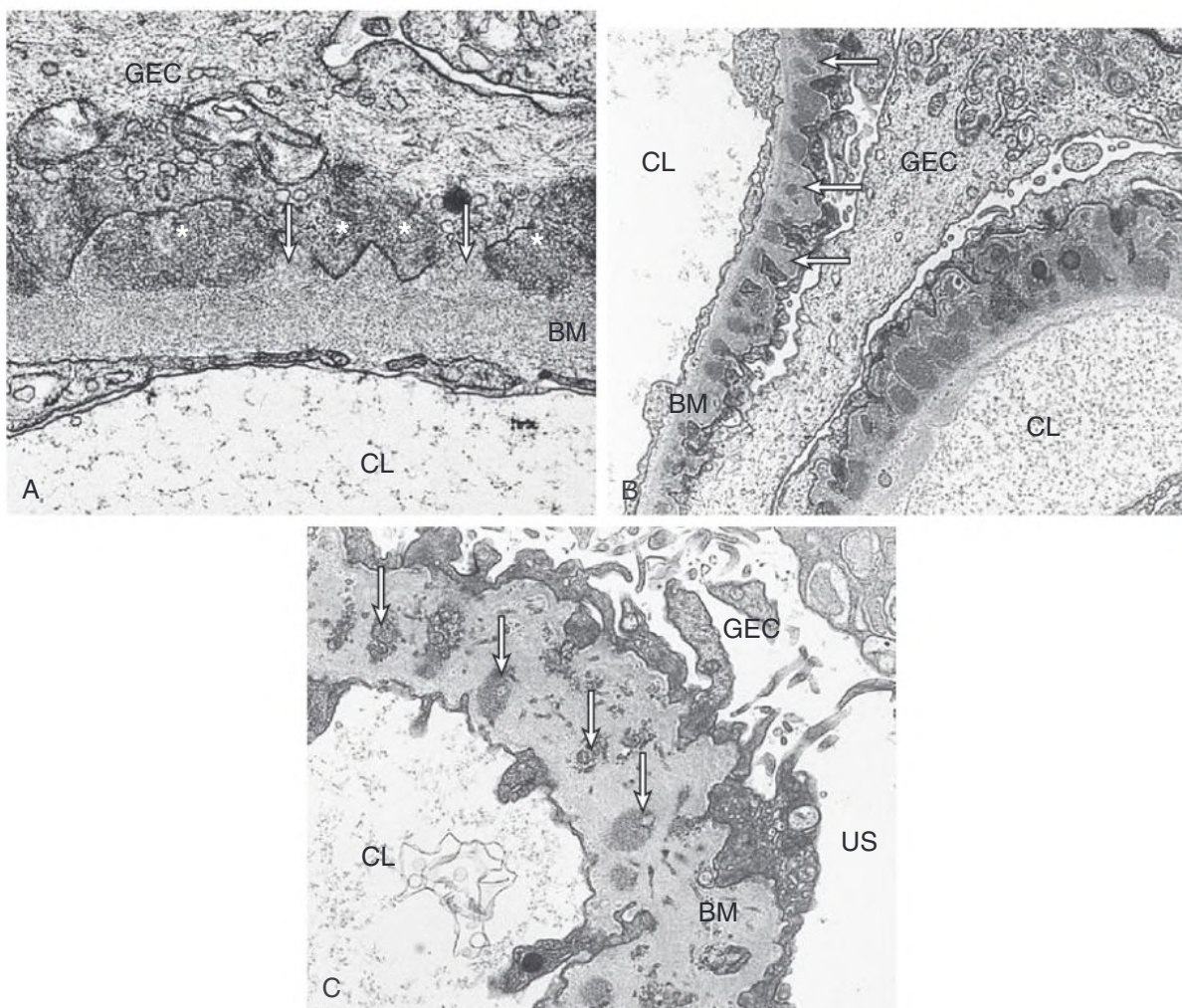


Fig. 21.6 Electron Microscopy in Membranous Nephropathy (MN). (A) Early (stage II) MN. Glomerular capillary wall with discrete, electron-dense deposits on the subepithelial surface of the basement membrane (BM) corresponding to granular deposits of IgG detected by immunofluorescence microscopy (corresponding to light micrograph in B). There are diffuse, granular immune complex deposits (white asterisks) along outer surface of the capillary wall, with effacement of overlying podocyte foot processes. Small extensions of BM between deposits (arrows) are also evident and represent the projections that are seen as spikes by light microscopy with silver methenamine staining. (B) More advanced (stage III) MN. Two glomerular capillary loops show involvement of the BM by the immune complex deposition (arrows). There is prominent membrane synthesis surrounding and incorporating these deposits into the BM (corresponding to spikes seen on silver-stained histologic preparations). Overlying cells continue to demonstrate widespread effacement of foot processes. (C) Morphologically advanced (stage IV) MN. Capillary BM is diffusely thickened; scattered electron-dense immune deposits (arrows) are present throughout its thickness, in addition to scattered subepithelial deposits. Overlying GECs continue to demonstrate effacement of foot processes. CL, Capillary lumen; GEC, glomerular epithelial cell; US, urinary space. (Original magnification $\times 18,000$.) (Courtesy C.E. Alpers.)

examination, appropriate laboratory evaluation should include a complement profile, antinuclear antibodies, hepatitis and human immunodeficiency virus (HIV) serology, chest radiography, stool testing for occult blood, mammography in women, and prostate-specific antigen testing with digital rectal examination in men. In women aged 20 to 50 years, a high index of suspicion is warranted for underlying lupus. This diagnosis can be particularly difficult to make because most of these patients have no systemic symptoms, and serologic markers of systemic lupus erythematosus are often absent. Membranous lupus accounts for 8% to 27% of cases of lupus nephritis (see [Chapter 27](#)).

In adults, regardless of age, malignancy is an important secondary cause of MN (see [Table 21.1](#)) and in some patients, the tumor may

not be evident at presentation. Although the diagnosis of anti-PLA₂R-associated MN renders an associated underlying malignancy unlikely,³⁴ given the therapeutic implications regarding immunosuppression affecting cancer growth and therapy, each case should still be carefully assessed and evaluations beyond routine age-appropriate testing considered if patients have risk factors for cancer. All anti-PLA₂R-negative cases merit a more thorough search for a malignant tumor.^{14,35}

HBV-associated MN is also a common secondary cause in countries where HBV is endemic and can occur with or without overt liver disease. It can affect both adults and children who are chronic carriers of HBV (positive HBsAg, HBcAg, and usually HBeAg). The presence of anti-PLA₂R-antibodies is well documented in HBV-associated MN.^{9,36}

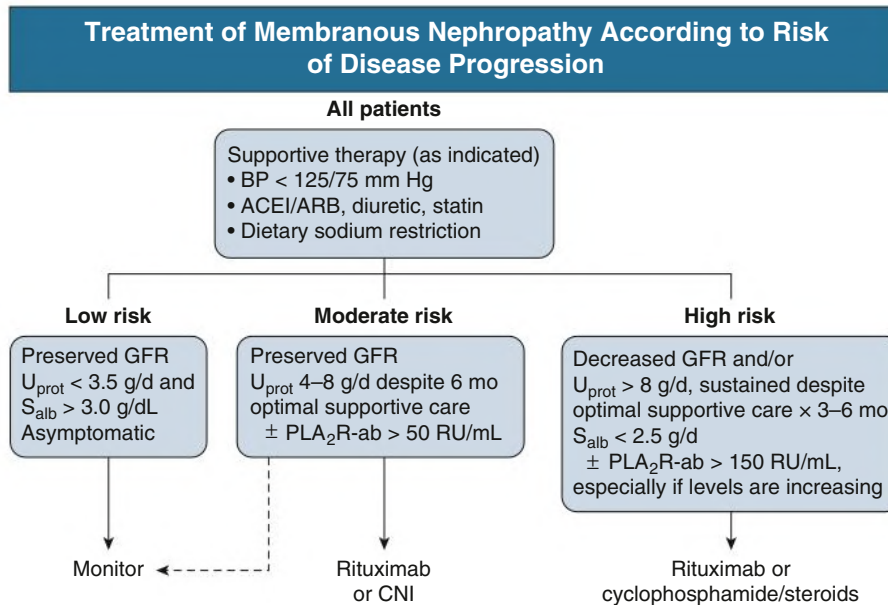


Fig. 21.7 Algorithm for Treatment of Membranous Nephropathy According to Risk of Disease Progression. All patients should receive supportive therapy. Low-risk patients can continue to be monitored on supportive therapy but should be periodically reassessed if they become symptomatic or their clinical parameters worsen. Close monitoring may also be appropriate for moderate-risk patients (*dashed arrow*), especially if they exhibit declining titers of anti-PLA₂R. Rituximab or calcineurin inhibitors (CNIs) are appropriate for moderate-risk patients, whereas higher-risk patients may require cytotoxic agents such as cyclophosphamide and corticosteroids, or extended durations of rituximab. Very high-risk patients (not shown), such as those with life-threatening complications of nephrotic syndrome or rapidly deteriorating kidney function, may warrant cytotoxic agents as first-line therapy. There are no threshold anti-PLA₂R levels to dictate initiation of immunosuppressive treatment but, if present, they should be followed to see if they are increasing or decreasing before starting or modifying therapy. ACEI, Angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BP, blood pressure; GFR, glomerular filtration rate; PLA₂R-ab, anti-PLA₂R antibodies; Salb, serum albumin (g/dL); U_{prot}, urine protein in grams/day.

In children, HBV-associated MN most often presents as the nephrotic syndrome and usually follows a benign course.³⁷ In adults, progressive loss of GFR is a more common outcome, and a controlled trial to assess the utility of combined therapy with antiviral therapy and tacrolimus has been proposed (<https://www.clinicaltrials.gov>, NCT03062813).

MN secondary to drugs usually resolves after discontinuation of the offending agent. The time to resolution, however, varies significantly from as early as 1 week for the drugs more commonly implicated (e.g., for nonsteroidal antiinflammatory drugs and proton pump inhibitors), compared with several years for the more classic but rarely used drugs such as gold and D-penicillamine. Many other kidney disorders have been seen in association with or superimposed on MN, including IgA nephropathy, FSGS, anti-GBM disease, ANCA vasculitis, acute interstitial nephritis, and diabetic nephropathy.

NATURAL HISTORY AND PROGNOSIS

The clinical course of MN varies widely. Spontaneous remissions in proteinuria occur in up to 30% of patients. Female gender and lower-grade (nephrotic) proteinuria at presentation are the only two features associated with a higher likelihood of spontaneous remission.³⁸ Inclusion of such patients with nephrotic-range proteinuria in natural history studies is likely to have produced a bias in kidney survival. For example, one study of untreated patients reported 72% kidney survival at 8 years, although more than one-third did not have nephrotic-range proteinuria at presentation and more than 50% had less than 5 g/day.³⁹ Despite this, the rate of end-stage kidney disease (ESKD) was 25% at 8 years and almost 50% by 15 years. Patients presenting with

less than 3.5 g/day of proteinuria, no RBC casts, no hypertension, normal kidney function, and no systemic features suggestive of a secondary cause have a generally favorable prognosis. However, these patients must be monitored, because up to 50% will develop nephrotic-range proteinuria at some time in the disease course, often within 2 years after presentation.

Traditional clinical variables including sustained proteinuria and kidney function are associated with long-term kidney outcome and can be used to classify patients into low, moderate, or high risk for progression to kidney failure (Fig. 21.7). One validated model to calculate risk in MN takes into consideration the initial creatinine clearance, the slope of the creatinine clearance during a fixed period, and the lowest level of proteinuria during that observation period.⁴⁰ The model predicts that patients with a normal creatinine clearance at presentation that remains stable during a 6-month period and persistent proteinuria of less than 4 g/24 h have a less than 5% chance of progression, and only conservative treatment is recommended. In contrast, patients with proteinuria of 4 to 8 g/24 h during the same time frame have a 55% probability for development of kidney impairment, and those with persistent proteinuria greater than 8 g/24 h have a 66% to 80% probability of progression to chronic kidney impairment within 10 years.

Anti-PLA₂R-antibody titer at the time of diagnosis can assist in prognostication (Table 21.4), although serum anti-PLA₂R titers have yet to be fully incorporated into prediction models. Testing for circulating anti-PLA₂R antibody can detect increases in MN immunologic activity before changes in classic laboratory parameters become apparent^{1,41} (Fig. 21.8). The presence and titer of anti-PLA₂R antibody

TABLE 21.4 Risk Factors Associated With Loss of Kidney Function in Membranous Nephropathy

Factors	Predictor
Clinical Features	
Age	Older > younger
Sex	Male > female
Hypertension	Present
Nephrotic syndrome	Present
Traditional Laboratory Parameters	
Albumin	<1.5 g/dL
Creatinine (eGFR)	Above normal (reduced)
Proteinuria	Sustained high-grade proteinuria
Kidney Biopsy Features	
Glomerulosclerosis (focal or global)	Present
Interstitial fibrosis	Present
Autoantibodies	
Anti-PLA ₂ R titer	Highest tertile > lowest tertile
Epitope spreading	Spreading > No epitope spreading
Antibodies to intracellular antigens	Present
Treatment and Response	
Immunosuppression	Not used > used, if indicated
Response to treatment	No response > partial or complete remission

eGFR, Estimated glomerular filtration rate; PLA₂R, phospholipase A₂ receptor.

can help diagnose primary MN, predict who might have spontaneous remission, monitor disease activity and response to therapy, identify those at risk for progression, and, most significantly, decide when to minimize or stop treatment.⁴²

A low or decreasing serum anti-PLA₂R antibody titer is associated with a greater likelihood of spontaneous and complete remission,^{43,44} but with a lag time that reflects the time required to stop forming deposits and allow the podocyte to recover from the complement-mediated injury. Conversely, an initially high or increasing trajectory of anti-PLA₂R suggests that the clinical manifestations will continue and may worsen in the short-term. The anti-PLA₂R antibody titers cannot be regarded on their own as indication to postpone or initiate immunotherapy, as clinical parameters such as serum albumin and degree of proteinuria and kidney impairment also must factor into decisions.

Other factors relating to the repertoire of circulating autoantibodies have been proposed as negative prognostic factors. The presence of “epitope spreading,” that is, autoantibodies to portions of PLA₂R in addition to the immunodominant N-terminal epitope (see Fig. 21.3), has been associated with worse clinical outcomes in some studies⁴⁵ but not others.⁴⁶ The presence of additional autoantibodies to intracellular antigens has been suggested to independently convey worse prognosis.⁴⁷

Beyond anti-PLA₂R, other biomarkers, including urinary α_1 -microglobulin, β_2 -microglobulin, IgM, and IgG, are also associated with MN progression.^{48,49} These markers measured together at a single time point have a higher positive predictive value than proteinuria alone, but independent prediction value beyond clinical variables and anti-PLA₂R antibodies requires further exploration.

The severity of chronic changes seen on the biopsy specimen (i.e., degree of glomerulosclerosis, tubulointerstitial fibrosis, and vascular disease) is associated with a poor prognosis but more closely reflects initial GFR than the subsequent rate of decline in GFR.^{25,50,51} Other pathologic features, including the percentage of glomeruli with glomerulosclerosis and the configuration of the immune deposits (synchronic/single stage or heterogeneous/multistage) on EM, have also been suggested as predictors of outcome and response to treatment, although these features did not add independent prognostic value in a larger cohort study.²⁵

Importance of Complete or Partial Remission

Relapse from a complete remission (CR) occurs in approximately 25% to 40% of MN cases, but the timing is unpredictable. Relapses have been reported up to 20 years after the primary remission. However, most patients relapsing after CR only develop subnephrotic-range proteinuria and will maintain stable long-term kidney function with conservative management alone.⁵² In contrast, the relapse rate is as high as 50% in those achieving only a partial remission (PR).

Ten-year kidney survival in patients with CR is 100%; with PR, 90%; and with no remission, only 45%.⁵³ CR was defined by proteinuria of 0.3 g/day or less, and PR was defined by a proteinuria value less than 3.5 g/day plus a 50% reduction from its peak value. A recent update in this cohort suggested that the durability of CR or PR, whether drug induced or spontaneous, is closely related to the long-term outcome.⁵⁴ This suggests that CR and PR may become acceptable shorter-term surrogate endpoints for kidney failure for clinical trials rather than reduction in GFR, which commonly takes years to evolve in MN.⁵⁵

TREATMENT

Nonimmunosuppressive Therapy

Conservative management of MN is directed at control of edema, hypertension, hyperlipidemia, and proteinuria and is similar to that used for nephrotic syndrome of any etiology (see Chapter 16). For patients with proteinuria of greater than 1 g/day, the target for blood pressure is 125/75 mm Hg unless contraindicated for clinical reasons. Angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers (ARBs) can reduce proteinuria and slow progression of kidney disease (see Chapter 82). A meta-analysis of the largest kidney protection trials using ACE inhibitors showed that the degree of protection is closely correlated to the degree of proteinuria reduction. None of these studies has focused on the specific effect of renin-angiotensin system (RAS) blockade in MN. In secondary analyses, the number of patients with MN has been small, and although the use of ACE inhibitors has been associated with significant improvement in some series, their antiproteinuric effect was modest (<30% reduction in proteinuria) in others. When effective, the benefit of RAS blockade occurs early, usually within the first 3 months of initiation of treatment. Even patients at low risk for progression (proteinuria <4 g/24 h) should be treated with ACE inhibitors or ARBs because this may reduce proteinuria and offer additional kidney protection, with minimal risk of significant adverse effect. Patients must also follow a low-salt diet (1.5–2 g sodium/day) to achieve the maximum benefit from RAS blockade.

When the proteinuria is in the nephrotic range, there is a clear increase in cardiovascular risk, with a threefold to fivefold increase in both coronary events and death rates in this population.⁵⁶ Patients with significant proteinuria almost always have pronounced hyperlipidemia. Although not proven, we recommend the use of statins to reduce low-density lipoprotein cholesterol to less than 100 mg/dL (<2 mmol/L), especially if proteinuria remains in the nephrotic range (see Chapters 82 and 85). Both RAS blockade and lipid control should be initiated early in these nephrotic MN patients at higher risk of progression.

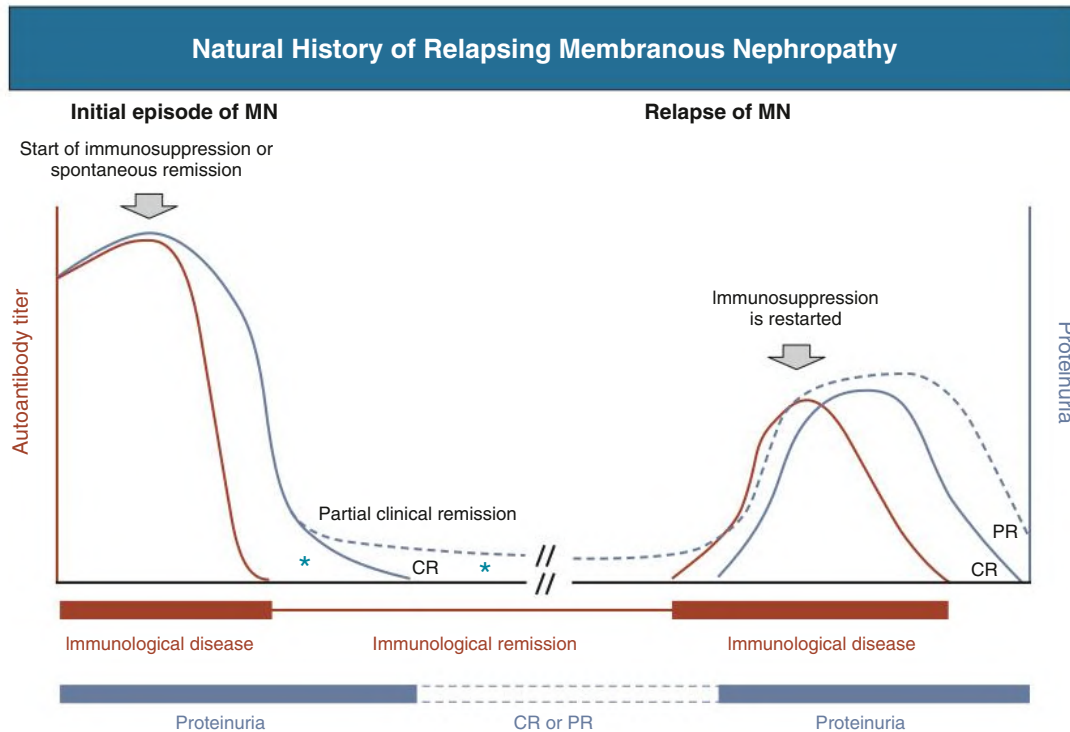


Fig. 21.8 Relationship of Immunologic and Clinical Courses of Disease in Membranous Nephropathy During Initial Disease and Relapse. Autoantibody (e.g., anti-PLA2R) titer is depicted in red and reflects the time course of immunologic (or serologic) disease. The start of the immunologic disease typically predates the clinical detection of MN by many months. Autoantibodies start to decline after successful immunosuppressive therapy or the onset of a spontaneous remission. Once the antibodies have disappeared, the patient is in an immunologic remission. The proteinuria (blue) lags behind the autoantibody curve and requires more time for the patient to achieve a partial remission (PR) and then a complete remission (CR). In some patients (dashed blue line), due to prolonged disease and more permanent damage to the kidney, levels of proteinuria may remain elevated in the subnephrotic range despite immunologic remission. Note that there are periods (asterisks) when proteinuria may still be present in the absence of detectable autoantibody. In 25% to 30% of patients, an immunologic and clinical relapse may occur, requiring another round of immunosuppression.

Persistent nephrotic syndrome, especially when caused by MN, is associated with an important risk of thromboembolic events.⁵⁷ Hypoalbuminemia appears to be a more robust predictor of venous thromboembolic events (VTEs) than the level of proteinuria. The risk of VTE increases dramatically if serum albumin is less than 2.5 g/dL.⁵⁸ These events tend to occur within the first 6 months of diagnosis but may occur up to 2 years from initial presentation. Patients with MN are also at significant risk of arterial thrombotic and cardiovascular events. Early in the disease course, these events occur at a rate that is commensurate to or exceeding that of ESKD.

Prophylactic anticoagulation seems beneficial in reducing fatal thromboembolic episodes in nephrotic patients with MN, without a concomitant increase in the risk of bleeding.⁵⁹ No randomized controlled trial (RCT) has ever been done, however, and thus there is no current consensus about prophylactic anticoagulation and no laboratory test that can predict with any accuracy such an event. Using VTE risk data informed by retrospective cohorts, and bleeding risk estimates based upon data from a large trial of warfarin in atrial fibrillation (ATRIA), a decision analysis balanced the risk of thrombosis versus bleed, considering risk thresholds for adverse events (<https://www.med.unc.edu/gntools/>).⁶⁰ The benefits of prophylactic anticoagulant therapy with warfarin appear to outweigh the risks in patients with serum albumin less than 2 or 2.5 g/dL in patients with a low to intermediate risk of bleeding, according to criteria outlined in the ATRIA study.⁶¹ An independent retrospective cohort supported use of an alternate regimen of anticoagulation with low-molecular-weight heparin in patients with nephrotic syndrome due to a spectrum of

glomerular pathologies.⁶² A shortcoming of the available data is the challenge of quantifying bleeding risk in patients with nephrotic syndrome, which can impact stability of anticoagulation with warfarin and some formulations of low-molecular-weight heparin. There are little data regarding safety of direct oral anticoagulants in this patient population.

Immunosuppressive Therapy

Several regimens using a variety of immunosuppressive agents can successfully reduce proteinuria in MN patients. Many unresolved questions are discussed in the Kidney Disease: Improving Global Outcomes (KDIGO) clinical practice guidelines for glomerulonephritis, including duration of conservative therapy while awaiting a spontaneous remission, when to initiate immunosuppressive therapy, the most effective and safest of the available agents, and duration of treatment before treatment is considered futile.^{62a}

Specific immunosuppressive drug therapy should not be considered unless the patient has had persistent nephrotic-range proteinuria (>4 g/day) and the proteinuria has not declined by more than 50% from baseline, over a minimum observation period of 6 months, despite maximum antihypertensive and antiproteinuric therapy. Other suggested criteria for early intervention are the presence of severe disabling or life-threatening symptoms related to the nephrotic syndrome or a rise in serum creatinine of more than 30% within 12 months, which would define a very high risk patient per the KDIGO guidelines. Worsening GFR should be documented as related to disease progression and not a complication, such as renal vein thrombosis,

transformation to crescentic MN variant, hemodynamic or volume-related changes from medications, or drug-induced interstitial nephritis. Low and/or declining titers of the PLA₂R antibody are associated with a higher likelihood of spontaneous remission, and very high levels of the anti-PLA₂R might be an indication for early intervention.^{8,41}

Corticosteroids

Three RCTs of corticosteroids in primary MN treatment showed no significant long-term beneficial effect on proteinuria, rate of disease progression, or kidney survival.^{63,64} The use of oral corticosteroids as a single agent for the treatment of MN is therefore not recommended. The one exception may be the Japanese population, in whom long-term observational studies have indicated improvement in both proteinuria and kidney function preservation with use of corticosteroids as monotherapy.⁶⁵

Cytotoxic Agents Combined With Corticosteroids

In patients at moderate risk of progression, a significant benefit has been described with a regimen beginning with methylprednisolone pulses 1 g intravenously for 3 days at the start of months 1, 3, and 5 followed by oral prednisone 0.5 mg/kg/day for 27 days, and each cycle followed by 1 month of treatment with an oral cytotoxic agent (cyclophosphamide or chlorambucil).⁶⁶ CR or PR was seen in almost 80% of treated patients, and both progression rate and kidney survival were significantly improved versus the supportive care group. Few adverse events were reported in this study, although relapses were seen within 2 years in 30% of the treatment group. Similar results were obtained in an RCT using the same cyclophosphamide-corticosteroid regimen to treat MN patients of Asian (East Indian) ethnicity.⁶⁷ Because the results of a cyclophosphamide-based regimen were similar to one based on chlorambucil, cyclophosphamide is most often used because of a better safety profile.

The safe use of cytotoxic agents in patients with deteriorating kidney function remains a challenge, even with appropriate adjustments in dose. One trial enrolled participants with a greater than 20% decline in GFR from baseline and demonstrated that the combination of corticosteroids and a cytotoxic drug (chlorambucil) offered marginally better protection against progressive kidney disease than placebo or cyclosporine but was associated with a high proportion of severe adverse events and a high study dropout rate.⁶⁸ Another study followed 65 high-risk patients with serum creatinine greater than 1.5 mg/dL treated with oral cyclophosphamide for 12 months plus corticosteroids.⁶⁹ Most patients achieved PR and had good kidney survival at 5 years, but two-thirds of participants had treatment-related complications such as bone marrow suppression and infections. Most adverse events could be managed by dose reduction, although some required permanent discontinuation of treatment.

In summary, cyclophosphamide used in combination with corticosteroids appears to be effective in the treatment of patients with nephrotic-range proteinuria due to primary MN, especially if GFR is well preserved at initiation of therapy. This combination may work even in those with impaired kidney function, but the supporting evidence is much less compelling, adverse effects are higher, and the likelihood of benefit is reduced, especially in patients with advanced kidney failure (GFR <30 mL/min/1.73m²).⁶⁹ The adverse effects of long-term cyclophosphamide therapy are the major drawbacks to the universal application of this treatment. These include increased susceptibility to infections, anemia, thrombocytopenia, nausea, vomiting, sterility, and, over time, malignant disease. Evidence from the MN literature suggests that the incidence of cancer is increased at a much lower level of exposure than previously considered; the number of malignant neoplasms increased with total cyclophosphamide exposure as low as 10 to 20 g

(~100 mg/day for 6 months).⁷⁰ Therefore, the use of cytotoxic agents should be limited to those patients with the highest risk of progression or with life-threatening complications of the nephrotic syndrome.

Calcineurin Inhibitors

Early uncontrolled studies using the calcineurin inhibitor (CNI) cyclosporine suggested an initial benefit but a high relapse rate. Cyclosporine may reduce proteinuria through its immunosuppressive effects but also by direct effects on the podocyte. In a single-blinded RCT, patients with corticosteroid-resistant MN were treated for 6 months with cyclosporine plus low-dose prednisone and compared with placebo plus prednisone.⁷¹ Cyclosporine was started at 3.5 mg/kg/day and titrated to reach a 12-hour trough level of 125 to 225 µg/L, with a mean trough level of approximately 150 µg/L ultimately achieved in the cyclosporine group. CR or PR was seen in 75% of cyclosporine-treated patients versus 22% of the controls. Cyclosporine was well tolerated. However, 38% relapsed within 6 months of discontinuation of treatment.

Only one RCT has used cyclosporine in patients with high-grade proteinuria and progressive decline in GFR, and both proteinuria and rate of kidney function loss were reduced with cyclosporine compared with placebo.⁷² This improvement in proteinuria was sustained for up to 2 years after cyclosporine was discontinued. In contrast to the UK study,⁶⁸ cyclosporine was introduced at a lower dose and slowly increased to minimize toxicity (treatment started at 3.5 mg/kg/day compared to 5 mg/kg/day in the UK study; target trough levels were similar). Treatment with longer-term cyclosporine (i.e., 12 months) has resulted in a higher rate of PR/CR (84%). Persistence of remission was maintained with doses of cyclosporine as low as 1 to 2 mg/kg, although relapses were still common if the cyclosporine level fell below 100 ng/mL.

Significant adverse effects seen with cyclosporine include hypertension, gingival hyperplasia, gastrointestinal complaints, muscle cramps, and, most important, nephrotoxicity. The latter depends on both dose and duration of treatment.

In a 12-month RCT (*n* = 48), monotherapy with tacrolimus was compared with a control group (conservative therapy only).⁷³ Proteinuria remission was 76% with tacrolimus versus 35% in the control group, and progression rate was also substantially slowed by the CNI. The relapse rate after stopping of the drug, however, approached 50% by the end of 2 years of follow-up.

In summary, both cyclosporine and tacrolimus reduce proteinuria in MN. Although relapses are common after short exposure (6–12 months), a longer duration and lower maintenance doses of CNI can maintain partial remission. No studies using CNIs have been of sufficient duration to confirm that maintenance of remission in proteinuria prolongs kidney survival. Side effects are substantial, with the major concern being nephrotoxicity, particularly if the CNI is not introduced at a low dose and slowly increased until an effective drug level is reached. The potential for nephrotoxicity with the CNIs increases if GFR is less than 40% of normal or rapidly deteriorating, or if there is a high degree of interstitial or vascular pathology accompanying the membranous lesion.

Mycophenolate Mofetil

Studies using mycophenolate mofetil (MMF) in MN patients report conflicting results. However, even in the most optimistic study, although initial response was high (used in combination with prednisone), the relapse rate within months approached 50%.⁷⁴ A comparison of MMF to a historical control group receiving cyclophosphamide showed no benefit of MMF in terms of effect or tolerability.⁷⁵ The addition of MMF to CNI therapy also does not appear to improve efficacy or relapse rate.⁷⁶

Rituximab

A favorable remission rate of 60% to 70% in patients treated with rituximab as a primary or salvage therapy prompted the design of larger RCTs to evaluate efficacy of this drug in MN.^{77,78} Furthermore, rituximab was shown to cause early reduction in anti-PLA₂R antibodies, a finding consistent with a favorable kidney prognosis.⁷⁹ Although the first RCT comparing rituximab plus conservative therapy to conservative therapy alone did not achieve a significant primary endpoint of remission at 6 months, anti-PLA₂R levels were significantly reduced by rituximab.⁸⁰ During the extended follow-up, remission rates after a mean of 17 months of observation increased to 65% in those treated with rituximab compared with 34% in those treated conservatively.

The largest RCT in primary MN ($n = 130$) was MENTOR (Membranous Nephropathy Trial of Rituximab), designed to test the hypothesis that rituximab was not inferior to cyclosporine for inducing and maintaining a complete or partial remission of proteinuria.^{81,82} Patients assigned to rituximab received 1 g on days 1 and 15. At 6 months, if proteinuria was reduced by more than 25% but there was no CR, a second course of rituximab was administered. In the cyclosporine group, trough levels of 125 to 175 ng/mL were targeted. If CR was observed at 6 months, cyclosporine was tapered and discontinued over 2 months. If proteinuria was reduced by more than 25%, cyclosporine was continued for an additional 6 months and then tapered and discontinued after 2 months. Sixty percent of patients in the rituximab group achieved the primary outcome PR/CR at 24 months versus only 20% in the cyclosporine group, supporting both noninferiority and statistical superiority for rituximab. At 12 months, 60% in the rituximab group and 52% in the cyclosporine group achieved PR/CR, demonstrating noninferiority for rituximab at this time point. The duration of action of rituximab in terms of B cell depletion may have extended the period of immunosuppression well beyond the 12-month treatment period. An important additional finding was a higher creatinine clearance at all time points in patients in remission within the rituximab group. In patients achieving remission, the anti-PLA₂R titer declined more rapidly in those who received rituximab. The rate of serious adverse events was 17% in the rituximab group and 31% in the cyclosporine group. Thus, the MENTOR study suggests that rituximab is at least as efficacious as cyclosporine at inducing remission, and is associated with a lower risk of relapse, with better preservation of creatinine clearance at 24 months, and a trend toward less serious adverse events.

The Sequential Treatment with Tacrolimus and Rituximab Versus Alternating Corticosteroids and Cyclophosphamide in Primary MN (STARMEN) trial evaluated the cyclical cyclophosphamide-corticosteroid regimen versus a combination of tacrolimus with rituximab added prior to tapering in order to potentially affect the rate of relapse after CNI withdrawal.⁸³ One group received the regimen with alternating months of corticosteroids and oral cyclophosphamide. The second group received tacrolimus targeted to achieve a trough level of 5 to 7 ng/mL for 6 months, followed by a single dose of 1 g of intravenous rituximab and a tacrolimus taper through the end of month 9. The primary outcome of PR/CR at 24 months was achieved in 84% in the corticosteroid-cyclophosphamide arm and 58% in the tacrolimus-rituximab arm. Patients treated with cyclophosphamide had a shorter time to PR/CR and immunologic remission and a trend toward better preservation of eGFR at 24 months. Patients in the cyclophosphamide group experienced more adverse events, including leukopenia and associated infections. Of note, two patients in the cyclophosphamide arm experienced reactivation of tuberculosis.

Although the STARMEN study provides important information, the short duration of tacrolimus therapy and the single dose of rituximab used in this trial do not allow clear comparison of either of

these agents to the known efficacy of standard alkylating agent-based therapy. The Rituximab or Cyclophosphamide in the Treatment of Membranous Nephropathy (RI-CYCLO) study directly compared the standard alternating corticosteroid-cyclophosphamide regimen to a single rituximab cycle (two 1 g doses given on days 1 and 15).⁸⁴ There was no significant difference in terms of PR/CR at 24 months between the groups, and adverse event rates were also similar. However, because longer-term adverse effects such as infertility or late malignancies, which could occur following treatment with cytotoxic agents, were not captured here, we still recommend consideration of rituximab as first-line treatment.

Complement Inhibition

Although products of complement activation are present in the glomeruli in a high fraction of cases of primary and secondary MN, presently there is no data from clinical trials to indicate if complement is a therapeutic target. An early 16-week RCT of the humanized anti-C5 monoclonal antibody, eculizumab, was terminated prematurely for lack of efficacy; however, there were concerns about adequate dosing because of proteinuria, and the results might be considered inconclusive. Given that complement inhibition would appear to be an appealing strategy during an early window of opportunity until effective immunosuppression lowers circulating autoantibody levels, various clinical trials have been initiated in patients with MN using LNP023 (iptacopan), a small molecule inhibitor of factor B of the alternative pathway; of OMS721 (narsoplimab), a neutralizing human monoclonal antibody targeting mannan-binding lectin-associated serine protease-2 (MASP-2) of the lectin pathway; and of APL-2 (pegcetacoplan), a small molecule C3 activation inhibitor.⁸⁵

Adrenocorticotrophic Hormone

Two small studies have reported the use of a synthetic form of adrenocorticotrophic hormone (ACTH) in MN. The exact mechanism of action of this agent in MN is unknown but likely unrelated to corticosteroid effects because corticosteroids alone are not beneficial in MN. An early dose escalation study of synthetic ACTH showed improved lipid profiles, with a small subgroup who received therapy for 1 year demonstrating sustained reduction in proteinuria.⁸⁶ In a small RCT, ACTH was compared with a standard cytotoxic plus corticosteroids regimen. The remission rate was the same in the two groups (80% vs. 90%), but the relapse rate in the ACTH group was lower (14% vs. 30%), after a follow-up of 1 year. Side effects of ACTH were few and included fluid retention, sleep disturbances, and bronze skin discoloration.⁸⁷

A dose escalation study found that natural ACTH gel significantly reduces proteinuria in proportion to drug exposure among most MN patients, and has an acceptable adverse event profile.⁸⁸ In contrast, a subsequent nonrandomized study using the synthetic agent found a substantially lower response rate and an incidence of adverse events approaching 95%, prompting the authors to suggest this agent not be used in MN.⁸⁹ Therefore, the most effective formulation of ACTH must be assessed in a larger RCT before it is recommended as standard therapy.

Treatment Approach

Achieving either CR or PR of the nephrotic syndrome is associated with prolonged kidney survival and a slower rate of kidney disease progression in patients with MN. Supportive or conservative care should be provided to all patients first, including diuretics, antihypertensive agents such as ACE inhibitors and ARBs, and lipid-lowering agents, with lifestyle modifications such as salt restriction and efforts toward smoking cessation (see [Chapter 82](#)).

Decisions regarding initiation of immunosuppression can be guided by stratifying patients into categories based on the risk of progression to kidney failure. Although persistent proteinuria is a key factor informing risk stratification, severe hypoalbuminemia places patients at significant risk of life-threatening complications of nephrotic syndrome and should be evaluated in parallel with proteinuria and factored into clinical decisions. Fig. 21.7 outlines a suggested management algorithm based on the level and persistence of proteinuria and incorporates information such as serum albumin and anti-PLA₂R antibody titer into risk stratification.

The traditional 6-month period of observation on optimized conservative therapy continues to be recommended. This might also include prophylactic anticoagulation in the setting of severe hypoalbuminemia and low bleeding risk, as described previously. In patients with high-grade proteinuria (>8 g/day), with hypoalbuminemia and persistently elevated or rising anti-PLA₂R antibody titer, initiation of immunotherapy can be considered after a shorter observation period (e.g., 3 months). Life-threatening complications of nephrotic syndrome such as thrombosis or worsening kidney function may also prompt earlier consideration of immunosuppression.

For patients at moderate risk of disease progression, evidence supports the use of either rituximab or a calcineurin inhibitor, although the MENTOR study supports a lower risk of relapse and better preservation of kidney function with rituximab. Cytotoxic/corticosteroid combinations are effective in moderate-risk or high-risk MN patients. This regimen, however, may be associated with important toxicity, and therefore is often reserved for patients at high risk of disease progression. Side effect profile, cost, and availability of these agents are important factors that will influence treatment selection for the individual patient.

Although beyond the scope of this discussion, bone protection and prophylaxis against *Pneumocystis jiroveci* pneumonia should be coadministered with corticosteroids and/or cytotoxic and rituximab therapy.

A reduction in anti-PLA₂R antibody titer will often occur prior to improvement in serum albumin or proteinuria, and the first evidence of clinical response may not be evident for 3 to 4 months. In the case of resistance to a treatment regimen or untenable adverse effects, an attempt at a second treatment regimen is often explored. Ideally, the approach should leave 2 to 3 months between treatment regimens to help immune system recovery.

Patients with severely reduced eGFR (<30 mL/min/1.73 m²) are less likely to benefit from immunosuppressive therapy (especially if the MN course has been slowly progressive and histopathology indicates significant interstitial fibrosis and glomerular obsolescence), and conservative therapy may be the best option for such patients.

FUTURE DIRECTIONS

The ongoing identification of novel MN antigens continues to highlight new mechanisms of disease development and will help elucidate additional biomarkers that can be used for prognostication and to assist in treatment decisions and monitoring of treatment response. Although great strides have been made in the design and execution of multicenter RCTs in this disease, the rate of adverse events in these trials has remained significant. Furthermore, the response rate remains modest, with 30% to 40% of study participants having no response to the allocated treatment regimen. This highlights the need to continue to search for personalized, novel, and safe treatment strategies for this important cause of kidney failure.

SELF-ASSESSMENT QUESTIONS

- Which statement on autoantigens in patients with membranous glomerulonephritis is *true*?
 - Phospholipase A2 receptor is the dominant autoantigen in Whites but not Asians.
 - THSD7A is a more common autoantigen in those of Japanese ancestry compared with Whites.
 - NELL1 is the autoantigen in most cases associated with malignancy.
 - Semaphorin 3B is the autoantigen in lupus-associated membranous glomerulonephritis.
 - Alpha-enolase is a membrane bound autoantigen, which is commonly shed into the glomerular basement membrane.
- Which statement on MN is *false*?
 - Primary MN has its peak incidence during the fourth and fifth decades of life.
 - Estimated annual incidence is 8 to 10 cases per 1 million population in Western countries.
 - The odds ratio of MN is almost 80 in individuals who are homozygous for particular HLA-DQA1 and PLA2R1 variants.
 - A majority of patients (70%–80%) with MN present with the nephrotic syndrome.
 - Microscopic hematuria is uncommon (5%–10%).
- Which statement on MN is *false*?
 - The predominant IgG subclass in many forms of primary MN is IgG3.
 - In earliest stages of MN, the glomeruli and interstitium appear normal on light microscopy.
 - In some patients with progressive disease, glomeruli exhibit areas that resemble secondary focal segmental glomerulosclerosis.
 - Granular glomerular capillary wall staining for IgG in MN is characteristic.
 - Phospholipase A2 receptor staining in the immune deposits exhibits a pattern that colocalizes with IgG.
- Which statement on the therapy of patients with membranous glomerulonephritis (MN) is *true*?
 - For patients with proteinuria of greater than 1 g/day, the target for blood pressure is 140/90 mm Hg.
 - Prophylactic anticoagulation should be given to all patients with MN.
 - Immunosuppressive drug therapy should not be considered unless the patient has had persistent nephrotic-range proteinuria.
 - Oral corticosteroids as a single agent for the treatment of MN are often effective.
 - Rituximab should not be first-line treatment because of longer-term adverse effects, such as late malignancies.

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